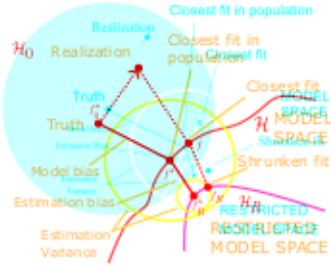
# Introduction to Machine Learning

## Regularization
## Bias-variance Tradeoff




**Learning goals**

- Understand the bias-variance trade-off
- Know the definition of model bias, estimation bias, and estimation variance

# BIAS-VARIANCE TRADEOFF

In this slide set, we will visualize the bias-variance trade-off.

We consider a DGP $\mathbb{P}_{xy}$ with $\mathcal{Y} \subset \mathbb{R}$ and the L2-loss $L$. We measure the distance between models $f : \mathcal{X} \to \mathbb{R}$ via

$$d(f, f') = \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\mathbf{x}}} \left[ L(f(\mathbf{x}), f'(\mathbf{x})) \right].$$

We define $f_0^*$ as the risk minimizer such that

$$f_0^* \in \arg\min_{f \in \mathcal{H}_0} \mathbb{E}_{(\mathbf{x},y) \sim \mathbb{P}_{xy}} \left[ L(y, f(\mathbf{x})) \right]$$

where $\mathcal{H}_0 = \{ f : \mathcal{X} \to \mathbb{R} \mid d(\underline{0}, f) < \infty \}$ and $\underline{0} : \mathcal{X} \to \{0\}$.

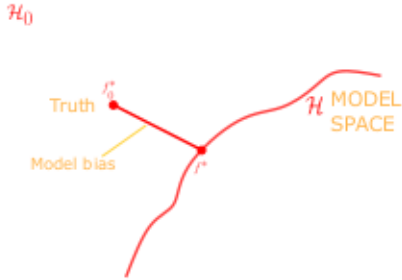
# BIAS-VARIANCE TRADEOFF / 2

Our model space $\mathcal{H}$ usually is a proper subset of $\mathcal{H}_0$ and in general $f_0^* \notin \mathcal{H}$.

We define $f^*$ as the risk minimizer in $\mathcal{H}$, i.e.,

$$f^* \in \arg\min_{f \in \mathcal{H}} \mathbb{E}_{(\mathbf{x},y)\sim\mathbb{P}_{xy}} \left[L(f(\mathbf{x}), y)\right].$$

$f^* \in \mathcal{H}$ is closest to $f_0^*$, and we call $d(f_0^*, f^*)$ the model bias.



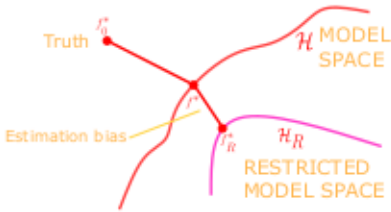
# BIAS-VARIANCE TRADEOFF / 3

By regularizing our model, we further restrict the model space so that $\mathcal{H}_R$ is a proper subset of $\mathcal{H}$. We define $f_R^*$ as the risk minimizer in $\mathcal{H}_R$, i.e.,

$$f_R^* \in \arg\min_{f \in \mathcal{H}_R} \mathbb{E}_{(\mathbf{x},y)\sim\mathbb{P}_{xy}} \left[L(f(\mathbf{x}, y)\right] .$$

model bias

$f_R^* \in \mathcal{H}_R$ is closest to $f_{\text{true}}$, and we call $d(f_R^*, f^*)$ the estimation bias.

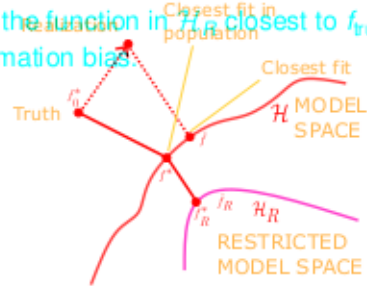


# BIAS-VARIANCE TRADEOFF / 4

We sample a finite dataset $\mathcal{D} = \left(\mathbf{x}^{(i)}, y^{(i)}\right)_{i=1}^{n} \in (\mathbb{R}^p \times \mathcal{Y})^n$ and find via ERM subset of $\mathcal{H}$. We define $f_R^*$ as the risk minimizer in $\mathcal{H}_R$, i.e.,

$$f_R^* \in \arg\min_{f \in \mathcal{H}_R} \mathbb{E}_{(\mathbf{x},y)\sim P_{xy}} \left[L\left(y, f(\mathbf{x})\right)\right].$$

It is the function in $\mathcal{H}_R$ closest to $f_{\text{true}}$, and we call $d(f_R^*, f^*)$ the estimation bias.




Note that the realization is only shown in the visualization for didactic purposes but is not an element of $\mathcal{H}_0$.

# BIAS-VARIANCE TRADEOFF / 5

Let's assume that $\hat{f}$ is an unbiased estimate of $f^*$ (e.g., valid for linear regression), and we repeat the sampling process of $\hat{f}$.

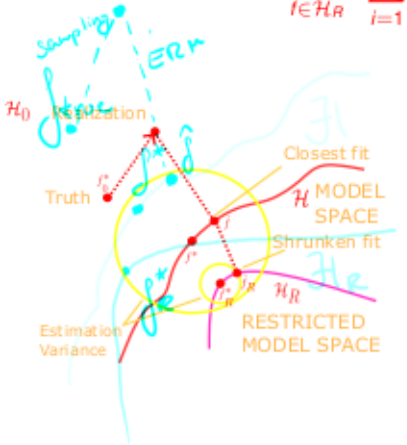
- We can measure the spread of sampled $\hat{f}$ around $f^*$ via $\delta = \mathrm{Var}_{\mathcal{D}}\left[d(f^*, \hat{f})\right]$ which we call the estimation variance.
- We visualize this as a circle around $f^*$ with radius $\delta$.

# BIAS-VARIANCE TRADEOFF / 6

We repeat the previous construction in the restricted model space $\mathcal{H}_R$ and sample $\hat{f}_R$ such that

$$\hat{f} \in \arg\min_{f \in \mathcal{H}} \sum_{i=1}^{n} L\left(y^{(i)}, \hat{f}(\mathbf{x}^{(i)})\right).$$

$$\hat{f}_R \in \arg\min_{f \in \mathcal{H}_R} \sum_{i=1}^{n} L\left(y^{(i)}, \hat{f}(\mathbf{x}^{(i)})\right).$$




Note:

- $L : \mathcal{Y} \times \mathbb{R}^g \to \mathbb{R}$ is overloaded.
- We can measure the spread of sampled $\hat{f}_R$ around $f_R^*$ via $\delta = \mathrm{var}_D\left[d(f_R^*, \hat{f}_R)\right]$ which we also call estimation variance.
- We observe that the increased bias results in a smaller estimation variance in $\mathcal{H}_R$ compared to $\mathcal{H}$.

# BIAS-VARIANCE TRADEOFF / 7

Let's assume that $\hat{f}$ is an unbiased estimate of $f^*$ (e.g., valid for linear regression), and we repeat the sampling process of $\hat{f}$.

- We can measure the spread of sampled $\hat{f}$ around $f^*$ via $\delta = \mathrm{Var}_{\mathcal{D}}\left[d(f^*, \hat{f})\right]$ which we call the estimation variance.
- We visualize this as a circle around $f^*$ with radius $\delta$.

# BIAS-VARIANCE TRADEOFF / 8

We repeat the previous construction in the restricted model space $\mathcal{H}_R$ and sample $\hat{f}_R$ such that

$$\hat{f}_R \in \arg\min_{f \in \mathcal{H}_R} \sum_{i=1}^{n} L\left(y^{(i)}, \hat{f}(\mathbf{x}^{(i)})\right).$$

- We can measure the spread of sampled $\hat{f}_R$ around $f_R^*$ via $\delta = \mathrm{Var}_{\mathcal{D}}\left[d(f^*, \hat{f}_R)\right]$ which we also call estimation variance.
- We observe that the increased bias results in a smaller estimation variance in $\mathcal{H}_R$ compared to $\mathcal{H}$.